

UNCLASSIFIED



Army EW and CEMA Doctrine

112-BOLCEW03

UNCLASSIFIED

Concrete Experience



UNCLASSIFIED



- What is this item?
- What does it do?
- How is it relevant to Military Operations?



- What is this item?
- What does it do?
- How is it relevant to Military Operations?



- What is this item?
- What does it do?
- How is it relevant to Military Operations?

How are all these items related to each other?

Enabling Learning Outcome

UNCLASSIFIED



The Student will define CEMA and EW operations and identify their placement within Army doctrine, units and operations in a classroom environment, given a network computer workstation, training aids, and appropriate references. Students will demonstrate an understanding of the divisions of CEMA/Electromagnetic Warfare Doctrine; operations and force structure in accordance with JP 3-13.1, ADRP 6-0, FM 3-12, FM 6-0, ATP 3-36, while achieving a minimum of 70% on all Practical Exercises (PE) and end of module exam.

UNCLASSIFIED

Learning Step Activities



UNCLASSIFIED

- Define Cyberspace Electromagnetic Activities (CEMA)
- Define Electromagnetic Warfare (EW)
- Define EW Divisions
- Compare different EW to other CEMA Operations
- Identify Army CEMA Doctrine
- Identify CEMA and EW within Army Operations
- Identify Force Structure of Army CEMA at the Corps and Below echelon
- Identify Key CEMA Personnel
- Identify Key CEMA support to other staffs

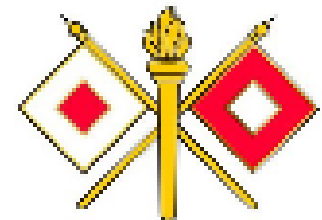
UNCLASSIFIED

Understanding CEMA

UNCLASSIFIED



- **Cyberspace Electromagnetic Activities** are actions leveraged to seize, retain, and exploit an advantage over adversaries and enemies in both cyberspace and the electromagnetic spectrum, while simultaneously denying and degrading adversary and enemy use of the same and protecting the mission command system.
- Commanders integrate and synchronize **cyber and electromagnetic** activities **across all** command **echelons** and **warfighting functions** as part of the operations process.
- Ensures information availability, protection, and delivery as well as the means to deny, degrade or disrupt the enemy's use of its command-and-control systems and other cyber capabilities.
- **CEMA** is Comprised of **Electromagnetic Warfare, Cyberspace Operations, Spectrum Management** operations.



UNCLASSIFIED

Electromagnetic Warfare Defined



UNCLASSIFIED

- **Electromagnetic Warfare** refers to military actions involving the use of electromagnetic and directed energy to control the Electromagnetic Spectrum or to attack the enemy. (FM 3-12)
- The purpose of electromagnetic warfare is to deny the enemy control of the Electromagnetic spectrum while ensuring freedom of maneuver for friendly operations in it.
- Electromagnetic Warfare is comprised of three divisions:
 - Electromagnetic Attack
 - Electromagnetic Protection
 - Electromagnetic Support

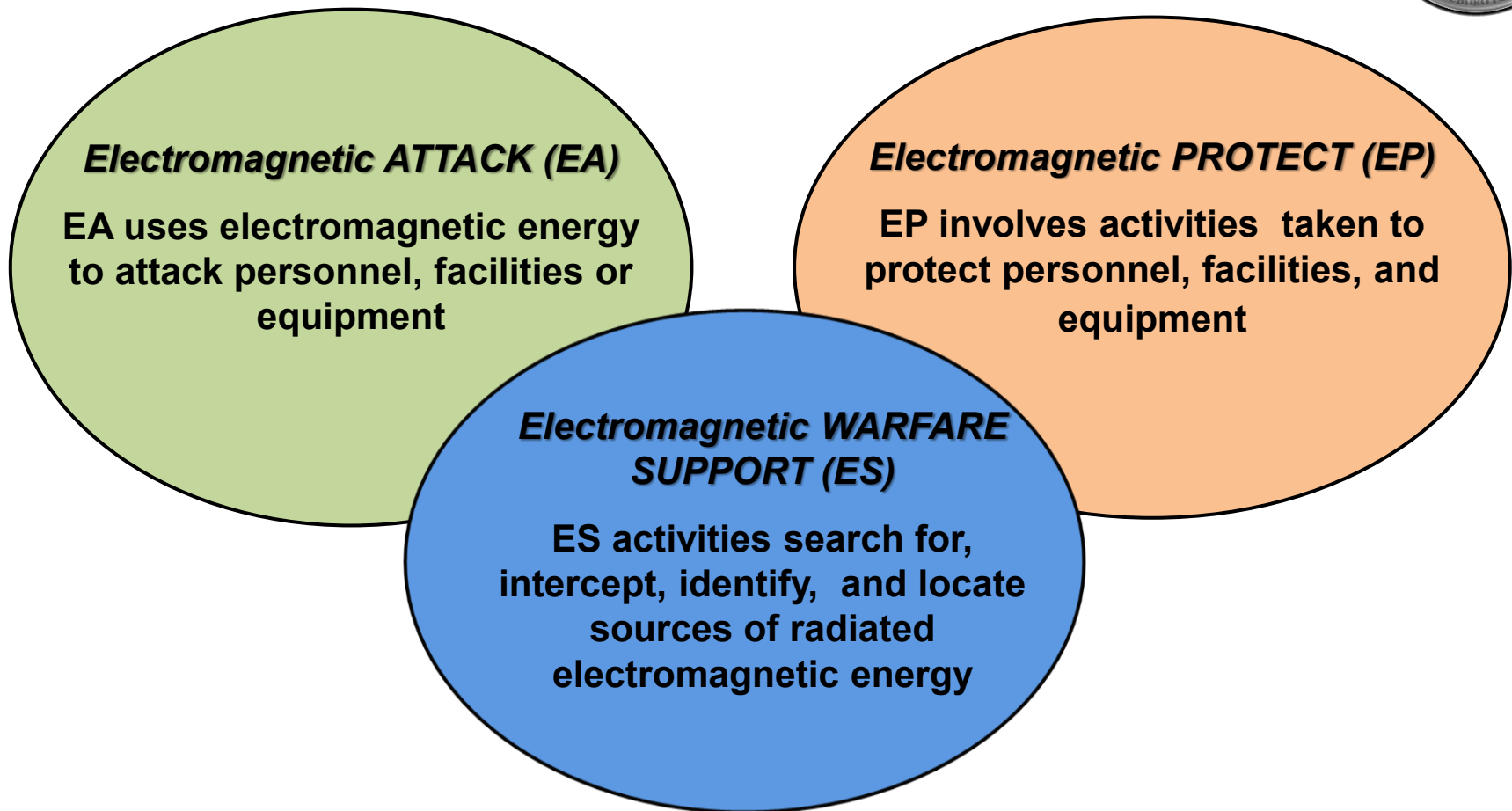


UNCLASSIFIED

Electromagnetic Warfare Divisions



UNCLASSIFIED



Each division of Electromagnetic warfare has its purpose and effects that support unified land operations (ULO).

UNCLASSIFIED

Electromagnetic Warfare



UNCLASSIFIED

Electromagnetic WARFARE

*Electromagnetic
WARFARE
SUPPORT (ES)*

*Electromagnetic
ATTACK (EA)*

*Electromagnetic
PROTECT (EP)*

Threat
Warning

Direction
Finding

Collection
Supporting EW

Non-Destructive

Destructive

EM Jamming

EM Deception

Chaff
Decoys

Directed
Energy

HARM

Protect from Friendly EW

Spectrum
Management

Protect from Enemy EW

Electromagnetic Emission Control
Hardening

UNCLASSIFIED

Comparison to Cyberspace Operations



UNCLASSIFIED

- Electromagnetic Warfare and Cyberspace Operations are both comprised of three categories.
- How are these categories different?
- How are they the same?
- Are cyberspace operations always distinct from Electromagnetic warfare operations?
- When can they be the same?
- When are they different?



Electromagnetic
Attack



Electromagnetic
Protection

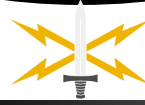


Electromagnetic
Support

Offensive
Cyberspace Ops



Defensive
Cyberspace Ops



DODIN OPs

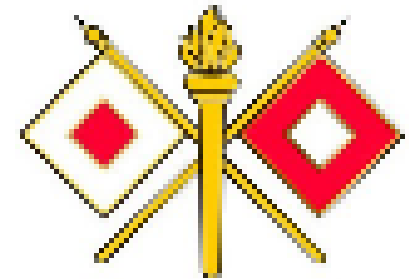


UNCLASSIFIED

Spectrum Management Operations



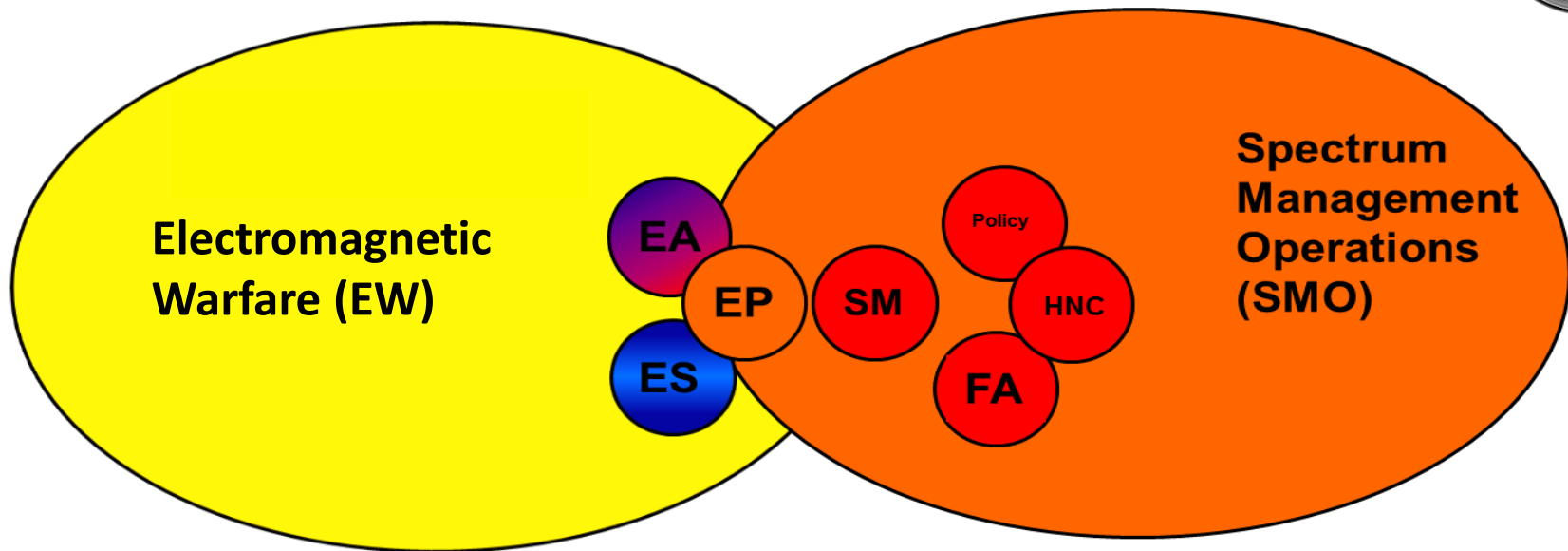
- Spectrum Management Operations (SMO) are interrelated functions of:
 - Spectrum Management
 - Frequency Assignment
 - Host Nation Coordination
 - Policy that enable planning, management, and execution of operations within electromagnetic operational environment during all phases of military operations
- Spectrum Management Operations is management portion of Electromagnetic Spectrum Operations (EMSO).
- EMSO is the combined effort of Spectrum management and Electromagnetic Warfare



Electromagnetic Spectrum Operations



UNCLASSIFIED



EA – Electronic Attack

EP – Electronic Protect

ES – Electronic Warfare Support

FA – Frequency Assessment

HNC – Host Nation Control

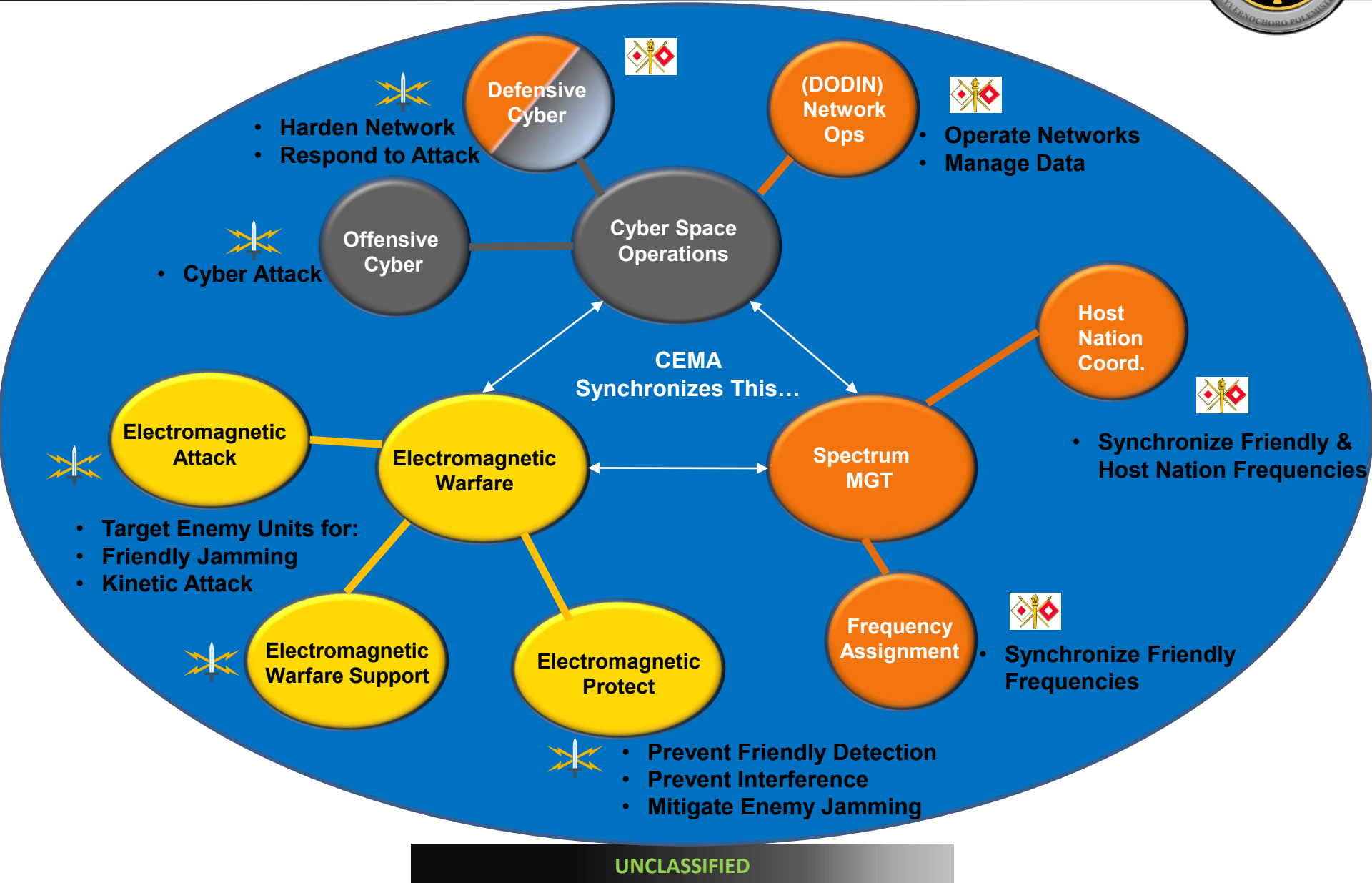
SM – Spectrum Management

- Electromagnetic spectrum operations (EMSO) is composed of EW and SMO
- The importance of the electromagnetic spectrum (EMS) and its relationship to the operational capabilities of the Army is the focus of EMSO

UNCLASSIFIED

Cyberspace and Electromagnetic Activities Combined

UNCLASSIFIED



UNCLASSIFIED

Important Publications for CEMA/EW



UNCLASSIFIED

- FM 3-12. *Army Cyberspace and Electromagnetic Warfare Operations.*
 - provides tactics and procedures for coordinating, integrating, and synchronizing Army CEMA operations.
- ATP 3-12.3. *Electromagnetic Warfare Techniques.*
 - provides doctrinal guidance and direction to the Army for conducting electronic warfare during unified land operations.
- ATP 3-12.4. *Electromagnetic Warfare Platoon*
 - provides doctrinal guidance for leaders responsible for planning, preparing, executing, and assessing operations of electromagnetic warfare platoons.
- ADP 6-0. *MISSION COMMAND: COMMAND AND CONTROL OF ARMY FORCES.*
 - Provides guidance and responsibilities of CEMA in support of Mission command.

UNCLASSIFIED

UNCLASSIFIED



CEMA/EW In Operations

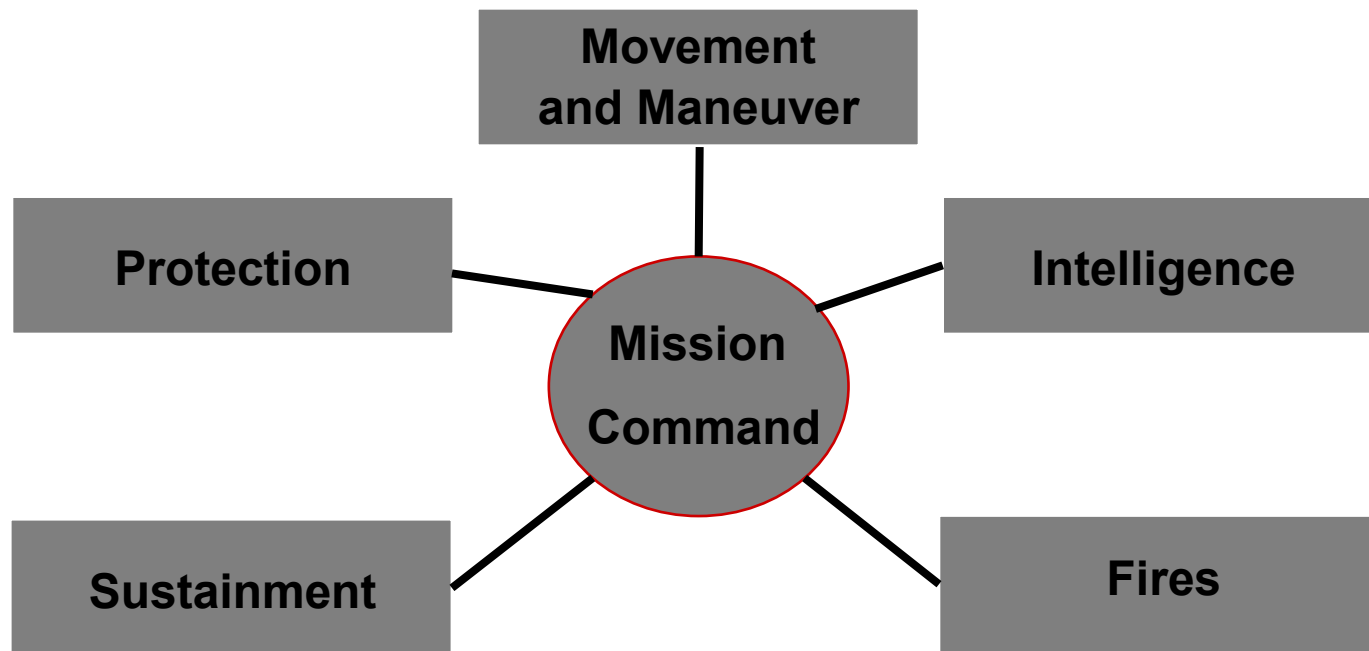
UNCLASSIFIED

Understanding Where CEMA/EW fits



UNCLASSIFIED

ARMY Warfighting Functions: A way to categorize and evaluate a unit's total combat power.



So where does CEMA, specifically EW, fall?

UNCLASSIFIED

Electromagnetic Warfare and Mission Command



UNCLASSIFIED

- Integration and synchronization of cyber electromagnetic activities (CEMA) is a task of this warfighting function. (ADRP 3-0)
- EW supports mission command warfighting function by:
 - Protecting mission command system from effects of friendly and adversary EW Operations
 - Controlling friendly EW systems through
 - Frequency deconfliction
 - Asset tracking
 - Employment execution
 - Reprogramming
 - Frequency registration with the spectrum manager

UNCLASSIFIED

Electromagnetic Warfare and Mission Command (cont.)

UNCLASSIFIED



- EW supports mission command warfighting function by (cont.):
 - Developing EW mission command tools to enhance required coordination between Army and Joint EW operations
 - Integrating, coordinating, deconflicting, and synchronizing EW operations through the CEMA working group
 - Increasing the commander's situational understanding through improved common operational picture input of EMS- and EW-related information
 - Monitoring and assessment EW operations

UNCLASSIFIED

Electromagnetic Warfare and Intelligence



UNCLASSIFIED

- EW enables the intelligence warfighting function by:
 - Increasing access for information collection assets by reducing anti-access, antipersonnel, and anti-systems threats
 - Increasing abilities to search for, intercept, identify, and locate sources of radiated EME in support of targeting, information tasks, and future operations
 - Increasing abilities in providing threat recognition and threat warning to the force
 - Providing indications and warning of threat emitters and radar
 - Denying and/or destroying threat information collection systems

UNCLASSIFIED

Electromagnetic Warfare and Fires



UNCLASSIFIED

- EW supports the fires warfighting function by:
 - Detecting and locating surface targets
 - Providing electro-optical-infrared and radio frequency countermeasures (EO-IR-CM)
 - Providing electromagnetic intrusion and deception
 - Jamming
 - Disrupting and/or degrading enemy sensors, C2 nodes, and infrastructure
 - Destroying targeted enemy systems

UNCLASSIFIED

Electromagnetic Warfare and Sustainment



UNCLASSIFIED

- EW supports the sustainment warfighting function by:
 - Protecting sustainment forces from friendly and adversary use of EW in static or mobile environments
 - Enhancing EME situational awareness through interception, detection, identification, and location of adversary EM emissions used to provide indications and warnings
 - Countering IEDs to support ground lines of communications
 - Providing spectrum deconfliction and EMCON procedures in support of sustainment control
 - Providing EMS obscuration, low-observability, and multispectral stealth

UNCLASSIFIED

Electromagnetic Warfare and Protection



UNCLASSIFIED

- EW enables the protection warfighting function by:
 - Enhancing EMS SA
 - Denying, disrupting, or destroying EM-triggered IEDs and enemy air defense systems
 - Deceiving enemy forces
 - Providing EMS obsuration, low-observability, and multispectral stealth

UNCLASSIFIED

Electromagnetic Warfare and Protection (cont.)



UNCLASSIFIED

- Providing EW countermeasures for platform survivability
 - Air and ground
- Protecting friendly personnel, equipment, and facilities from friendly and enemy EA, including friendly information systems and information
 - Includes coordination and use of both airborne and ground-based EA with higher and adjacent units

UNCLASSIFIED

So, what do CEMA/EW provide?



UNCLASSIFIED

Electronic Warfare Effects Planning Menu

EFFECT	Movement To Contact	Long Range Artillery	Deception	Air Assault	MISO Messages	Counter UAS	Cordon And Search	Wet Gap Crossing	React To Contact	Convoy Operations	Rear Area Security	USED FOR / AGAINST
KILL CHAIN COMMS												ENY C2 IN THE ATTACK
MONITOR												ENY COMMS/RADIATING DEVICES
BARRAGE JAMMING												ENY LISTENING DEVICES/SIGINT
FALSE TARGET												ENY RADAR
MESSAGING												ENY COMMS
SPOOFING												ENY COMMS/RADARS
ELECTRONIC PROBING												ENY COMMS/RADARS
COMMS DENIAL												ENY C2
RADAR DISRUPTION												RADARS
MILDEC / PSYOP / IO												ENY C2
ENY MESSAGE DENIAL												RADIO TV SYSTEMS
DATA LINK DISRUPTION												UAS/ C2/ FIRE CONTROL
TELEMETRY LINK DISRUPTION												UAS FLIGHT CONTROL FREQ.
ROUTE OVER WATCH												MONITOR EMS ALONG ROUTE
IMMINENT THREAT & WARNING												ENY FWD OBSERVERS
HERDING												ENY PACE PLAN
DECEPTION JAMMING												ENY SIGINT/RADAR
RC-IED												RC-IED DENY DET.
CONDITIONING												ENY COMMS

Technique / Effects	Meaning // How To Use
Kill chain COMMS	Disrupt ENY Communications during an attack
Monitor	Monitor spectrum activities- used during look windows (period of no jamming during operations)
Barrage jamming	Broad spectrum Jamming- used only during unique circumstances. Hi power across multiple frequencies of activity
False target	Injecting false targets (Aircraft) into the enemy Radar system to obscure true target or create false imprecision
Messaging	Jamming operation that plays selected messages instead of noise jamming
Spoofing	Radiating energy to mimic another system or systems in the spectrum to get a reaction from the enemy
Electronic probing	Deliberate jamming under controlled situations to gauge an enemy response and TTPs to jamming operations
Comms denial / degrade	Jamming of active signals within an area of operations to disrupt enemy communications at or above 90% / 50%
Radar disruption	Radar jamming can be done by: obscuration, false targets, or false position, disruption will use obscuration to deny enemy air picture
MILDEC / PSYOP / IO	Playing selected messages across selected frequencies at a targeted audience to include a desired effect
ENY message denial	Jamming operations that degrade enemy IO / propaganda being played across the spectrum
Data link disruption	Counter UAS jamming intended to disrupt the data or feed from a UAS from the ground control station --no flight parameters
Telemetry link disruption	Counter UAS jamming intended to disrupt the flight control of a UAS from the ground control station
Route over watch	Jamming operations that patrol along a convoy's route to provide KCC and deny detonation of RC-IEDs
Imminent threat & warning	Jamming operations that identify enemy communications by forward observers or spotters and disrupt those communications
Herding	Jamming operations intended to move an enemy to a more desirable comms system/frequency to enhance the collection effort
Deception jamming	Jamming operations intended to convince an enemy that friendly forces are going to do or intending to do a certain action
RC-IED	Deny detonation of RCIEDs along a route during friendly force movements
Conditioning	Requires substantial lead time, jamming operations to include complacency to jamming effects on systems or during certain times

TECHNIQUE / EFFECTS	MEANING // HOW TO USE
KILL CHAIN COMMS	Disrupt ENY Communications during an attack
MONITOR	Monitor spectrum activities- used during look windows (period of no jamming during operations)
BARRAGE JAMMING	Broad spectrum Jamming- used only during unique circumstances. Hi power across multiple frequencies of activity
FALSE TARGET	Injecting false targets (Aircraft) into the enemy Radar system to obscure true target or create false imprecision
MESSAGING	Jamming operation that play selected messages instead of noise jamming
SPOOFING	Radiating energy to mimic another system or systems in the spectrum to get a reaction from the enemy
ELECTRONIC PROBING	Deliberate jamming under controlled situations to gauge an enemy response and TTPs to jamming operations
COMMS DENIAL	Jamming or energy of active signals within an area of operations to disrupt enemy communications at or above 90%
RADAR DISRUPTION	Radar jamming can be done by: obscuration, false targets, or false position, disruption will use obscuration to deny enemy air picture
MILDEC / PSYOP / IO	Playing selected messages across selected frequencies at a targeted audience to include a desired effect
ENY MESSAGE DENIAL	Jamming operations that degrade enemy IO / propaganda being played across the spectrum
DATA LINK DISRUPTION	Counter UAS jamming intended to disrupt the data or feed from a UAS from the ground control station --no flight parameters
TELEMETRY LINK DISRUPTION	Counter UAS jamming intended to disrupt the flight control of a UAS from the ground control station
ROUTE OVER WATCH	Jamming operations that patrol along a convoy's route to provide KCC and I&W to ground forces
IMMINENT THREAT & WARNING	Jamming operations that identify enemy communications by forward observers or spotters and disrupt those communications
HERDING	Jamming operations intended to move an enemy PACE plan to more desirable system and frequencies for collection effort
DECEPTION JAMMING	Jamming operations intended to convince an enemy that friendly forces are going to do or intending to do a certain action
RC-IED	Deny det. Or RCIEDs along a route during friendly forces movements- time consuming and hard to measure effectiveness
CONDITIONING	Requires substantial lead time, jamming operations to include complacency to jamming effects on systems or during certain times

UNCLASSIFIED

UNCLASSIFIED



Organization of CEMA/EW at echelon

UNCLASSIFIED

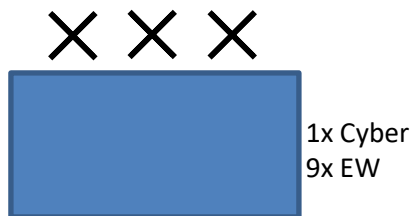
CEMA or EW?



UNCLASSIFIED

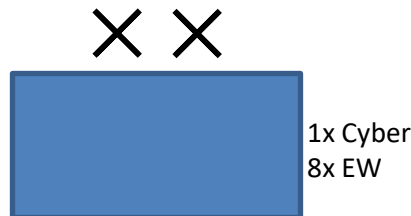
- Why are the terms CEMA and EW used so interchangeably?
 - Because almost all the organic CEMA capability (outside DODIN and Spectrum Management) at the CORPS and below echelon are EW related capabilities
 - Most of the CEMA personnel at the CORPS and below are EW (17E, 170B, 17B)
 - CEMA doctrine is relatively new.

CEMA Personnel Corps and Below



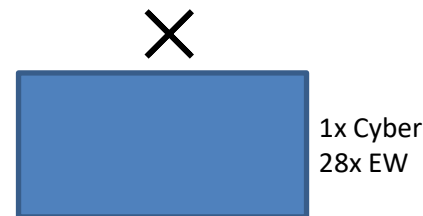
CEMA Cell

CEMA OIC (17B)
EWO (17B)
CWO (17A)
2x Tech (170B)
7x Enlisted EW Staff (17E)
1x Spectrum Manager (25E)



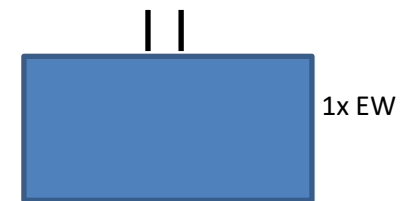
CEMA Cell

CEMA OIC (17B)
EWO (17B)
CWO (17A)
2x Tech (170B)
7x Enlisted EW Staff (17E)
1x Spectrum Manager (25E)



CEMA Cell

CEMA OIC (17B)
EWO (17B)
CWO (17A)
Tech (170B)
7x Enlisted EW Staff (17E)
1x Spectrum Manager (25E)



EW NCO (17E)

EW PLT
EW PL (17B)
5x EW NCOs (17E)
14x Enlisted EW (17E)

UNCLASSIFIED

CEMA Operations at Corps



UNCLASSIFIED

- Army forces at echelons corps level and below plan, integrate, and synch all aspects of cyberspace and EW operations with all WFF to create complementary and reinforcing effects.
- Based on its various roles, corps will coordinate directly or indirectly with USCYBERCOM cyber support element and CCMD joint cyberspace center.
- Commanders and staff elements execute CEMA to support the commanders' objectives
- Cyberspace requests to Corps will use the CERF.

UNCLASSIFIED

DIV to BDE-Level CEMA



UNCLASSIFIED

- These tactical echelons employ can employ organic EW capabilities through the CEMA cell/BDE EW PLT. These echelons also conducts DODIN ops through the S6/G6 staff.
- Some DCO-IDM capabilities reside at these echelons through the S-6/G6 staff, and the
- CEMA with S6/G6 staff coordinates and synchronizes for nonorganic DCO-IDM support.
- All OCO support is external and is coordinated through the CEMA cell.
- Staff may integrate nonorganic cyberspace and EW forces as authorized and required.

(See FM 3-94 for additional information regarding the corps to brigade roles and responsibilities.)

UNCLASSIFIED

CEMA Section



UNCLASSIFIED

- CEMA section of G-3 (S-3) from BDE to Corps coordinates and synchronizes operations for effective collaboration across staff elements.
- CEMA section participates in planning and targeting process and leads CEMA working group to support MDMP. Cyberspace planner requests for effects provided by non-organic resources.

Example Task Organization of CEMA Cell

CEWO

- Overall responsible for MDMP
- Gives guidance and intent
- Sets dead lines

FUOPS: Responsible for all planning

-**CWO:** Responsible for planning Cyber Effects. Serves as Liaison to Division when needed

-**EW O:** Responsible for integrating EW into BDE mission. Serves as primary plans officer

-**EW Tech:** Serves as technical advisor for planning. In charge of CEMA target nominations.

-**Spectrum Manager:** Responsible for clearing and submitting EW effects.

CUOPS: Responsible for tracking requests/reports and submitting RFIS

-**CEMA NCOIC:** Responsible for managing CUOPS and keeping running estimates current.

-**2XEW NCO:** Responsible for tracking reports and submitting and tracking RFIs

-**3xEW Soldiers:** Assists EW NCO as needed

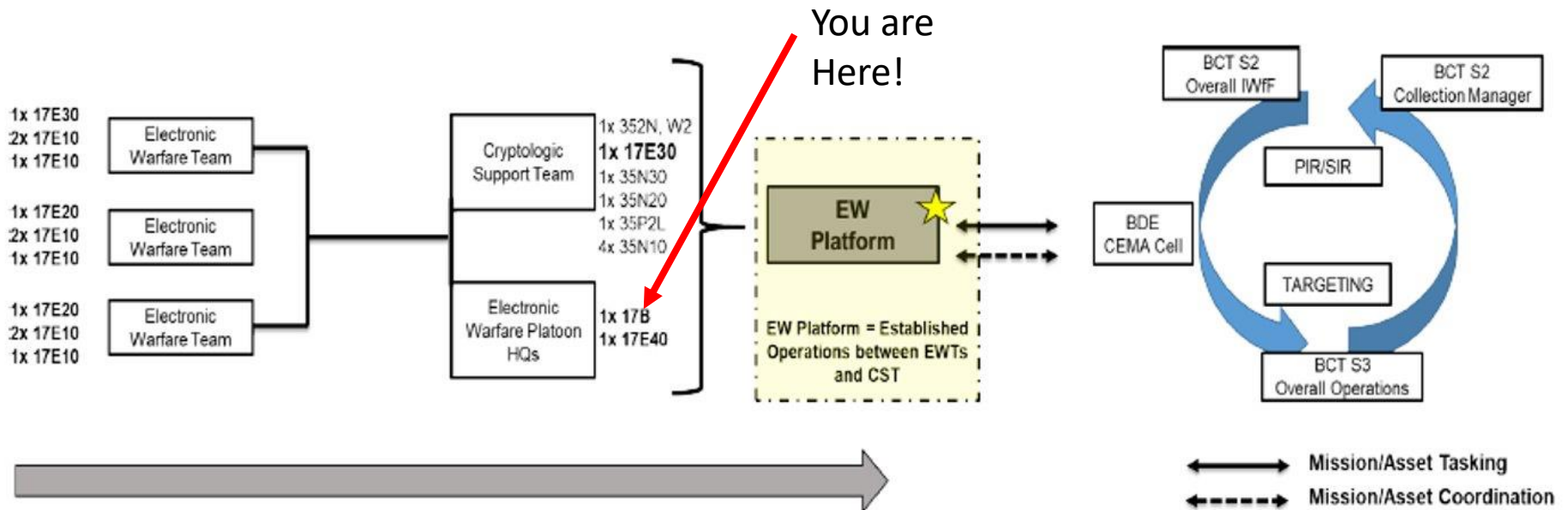
UNCLASSIFIED

BDE EW PLT



UNCLASSIFIED

- The EW platoon's mission is to conduct electromagnetic sensing and effects delivery.
- EW Platoons support operations to answer the commander's PIRs and provide early warning to protect the force.
- Commanders specifically assign EW platoon missions for any given operation based on known and projected friendly operations, the range of threats, and their understanding of potential areas of operations.



UNCLASSIFIED

CEMA at Battalion and below



UNCLASSIFIED

- Battalions rely on their brigade for core services, network accessibility, and network defense. Battalion S-6 performs planning and operations associated with the main and tactical command posts, including establishing connectivity with adjacent, subordinate, and higher elements.
- Battalions do have capabilities to support cybersecurity policies and request for information regarding cyberspace and the EMS.
- Companies rely on their battalion for network service, access, and network defense.
- Company performs planning and operations associated with the command post, including establishing connectivity with adjacent, subordinate, and higher elements. Commanders at this echelon are responsible for applicable cybersecurity measures.

UNCLASSIFIED

Additional Personnel that Support CEMA



UNCLASSIFIED

- G/S-6 Signal
- S-6 Spectrum Manager
- G/S-2 Intelligence
- IO Officer
- Fire Support
- SJA

UNCLASSIFIED



CEMA Support to other staff

Information Operations (IO)



UNCLASSIFIED

- Information operations is the integrated employment, during military operations, of information related capabilities (IRC) in concert with other lines of operation to influence, disrupt, corrupt, or usurp the decision making of enemies and adversaries while protecting our own (JP 3-13).
- Because cyberspace is fully part of the information environment, cyberspace operations are inherently IRCs.

UNCLASSIFIED

Intelligence Operations



UNCLASSIFIED

- Intelligence supports cyberspace operations through the application of the intelligence process, IPB, and information collection. Intelligence at all echelons supports DODIN operations, DCO, and OCO planning and assists with defining measures of performance and measures of effectiveness.
- G/S-2 coordinates with cyber planners and the G-6 (S-6 to use cyberspace capabilities for:
 - Enabling collection assets
 - Linking intelligence capabilities
 - Providing information capabilities
 - Providing near real time analysis
 - Information collection

UNCLASSIFIED

Space Operations



UNCLASSIFIED

- The relationship between the space and cyberspace domains is unique. Space operations depend on EMS for transport of information and control of space assets. Space provides a key global connectivity capability for cyberspace operations. Conversely, cyberspace operations provide a capability to execute space operations. This interrelationship is an important consideration across cyberspace operations, and particularly when conducting targeting in cyberspace.
- Space and cyberspace operations complement each other, but also have mutually exclusive mission sets and processes. Space provides and supports global access to cyberspace operations through satellite communications.

UNCLASSIFIED

Targeting



UNCLASSIFIED

- Targeting is means by which commanders and staffs pair targets with effects in and through cyberspace. Through MDMP, staff identifies high-value targets and high pay-off targets, and through CEMA working group identifies targets that may qualify for targeting using effects from cyberspace or EW operations.
 - Common operational picture, DCO, cyberspace security information, information collection efforts, and desired operational outcomes also guide targeting.
 - Army forces plan and execute (with proper authorities) effects on, in and through cyberspace and the EMS using available capabilities.
 - Units synch effects with the next higher command.

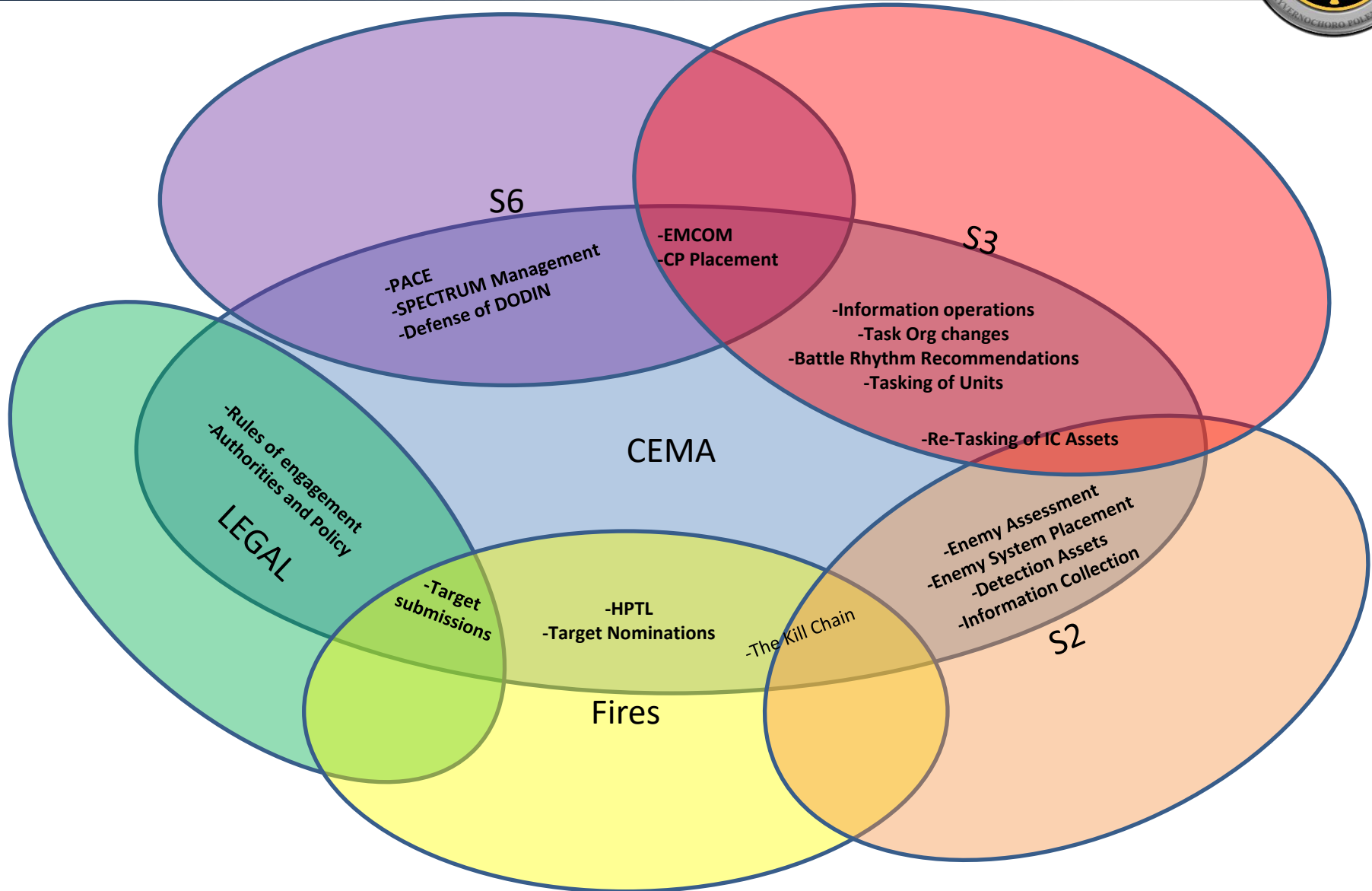
(CERF and JTASR/EARF are used to request cyberspace and EMS effects).

UNCLASSIFIED

CEMA/EW Integration With Staff



UNCLASSIFIED



UNCLASSIFIED

Practical Exercise



UNCLASSIFIED

- **Objective:** Define key personnel within CEMA
- **Material:** FM 3-12 and course slides
- **Tasks:**
 - Form teams of three students, research, build, and present a slide with the role and responsibilities of:
 - Cyber Electromagnetic Warfare Officer (CEWO)
 - Electromagnetic Warfare Officer
 - Cyber Warfare Officer
 - Electromagnetic Warfare Technician
 - CEMA/EW NCOIC
 - CEMA Spectrum Manager
 - EW Platoon Leader

UNCLASSIFIED

Summary



UNCLASSIFIED

- Define Cyberspace Electromagnetic Activities
- Define Electromagnetic Warfare
- Define Electromagnetic Warfare Divisions
- Compare different CEMA Operations
- Identify Army CEMA Doctrine
- Identify CEMA and EW within Army Operations
- Identify Force Structure of Army CEMA at the Corps and Below echelon
- Identify Key personnel within CEMA
- Identify Key CEMA support to other staffs

UNCLASSIFIED

UNCLASSIFIED



Application and Check on Learning

UNCLASSIFIED

References



UNCLASSIFIED

- FM 3-12. *Army Cyberspace and Electromagnetic Warfare Operations*. Apr 2017.
- ATP 3-36. *Electromagnetic Warfare Techniques*. Dec 2014.
- ADP 3-0. *Operations*. Oct 2017.
- ADRP 3-0. *Operations*. Oct 2017.
- ADP 5-0. *The Operations Process*. May 2012.
- ADRP 5-0. *The Operations Process*. May 2012.
- ADP 6-0. *Mission Command*. May 2012.
- ADRP 6-0. *Mission Command*. May 2012.

UNCLASSIFIED

Questions?



UNCLASSIFIED



UNCLASSIFIED